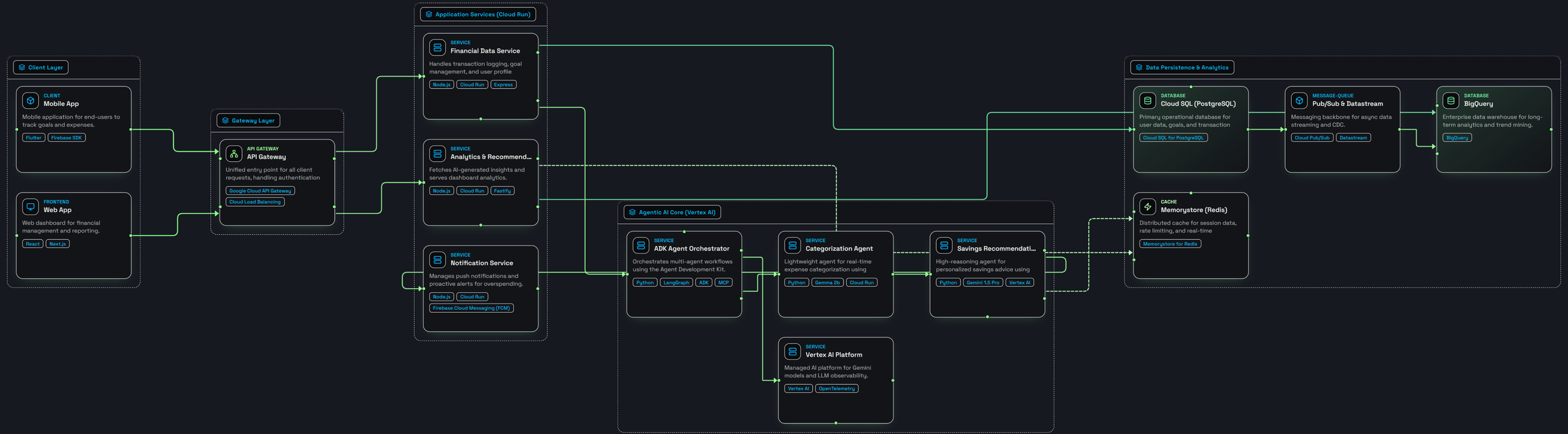
Client Layer
CLIENT
Mobile App
Mobile application for end-users to track goals and expenses.
Flutter
Firebase SDK
FRONTEND
Web App
Web dashboard for financial management and reporting.
React
Next.js
Gateway Layer
API GATEWAY
API Gateway
Unified entry point for all client requests, handling authentication
Google Cloud API Gateway
Cloud Load Balancing
Application Services (Cloud Run)
SERVICE
Financial Data Service
Handles transaction logging, goal management, and user profile
Node.js
Cloud Run
Express
SERVICE
Analytics & Recommend...
Fetches AI-generated insights and serves dashboard analytics.
Node.js
Cloud Run
Fastify
SERVICE
Notification Service
Manages push notifications and proactive alerts for overspending.
Node.js
Cloud Run
Firebase Cloud Messaging (FCM)
Agentic AI Core (Vertex AI)
SERVICE
ADK Agent Orchestrator
Orchestrates multi-agent workflows using the Agent Development Kit.
Python
LangGraph
ADK
MCP
SERVICE
Categorization Agent
Lightweight agent for real-time expense categorization using
Python
Gemma 2b
Cloud Run
SERVICE
Savings Recommendati...
High-reasoning agent for personalized savings advice using
Python
Gemini 1.5 Pro
Vertex AI
SERVICE
Vertex AI Platform
Managed AI platform for Gemini models and LLM observability.
Vertex AI
OpenTelemetry
Data Persistence & Analytics
DATABASE
Cloud SQL (PostgreSQL)
Primary operational database for user data, goals, and transaction
Cloud SQL for PostgreSQL
MESSAGE-QUEUE
Pub/Sub & Datastream
Messaging backbone for async data streaming and CDC.
Cloud Pub/Sub
Datastream
DATABASE
BigQuery
Enterprise data warehouse for long-term analytics and trend mining.
BigQuery
CACHE
Memorystore (Redis)
Distributed cache for session data, rate limiting, and real-time
Memorystore for Redis